

STPM PHYSICS • SEMESTER 3 • CHAPTER 24

Quantum Physics

24.1 Photon

DEFINITIONS & LAWS

- Photon** is a quantum (packet) of electromagnetic radiation of energy $E = hf$.
- Photoelectric effect** is the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency falls on it.
- Work function ϕ** is the minimum energy required to remove an electron from the surface of a metal; **threshold frequency f_0** is the minimum frequency for emission, $f_0 = \frac{\phi}{h}$.
- Stopping potential V_s** is the minimum reverse potential difference that reduces the photocurrent to zero; $eV_s = K_{\max}$.
- Einstein's photoelectric equation:** $hf = \phi + K_{\max}$, an electron absorbs one photon whose energy is shared between the work function and the kinetic energy.
- Observations that wave theory cannot explain: a threshold frequency exists; $K_{\max}$ is independent of intensity and depends on frequency; emission is instantaneous.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Photon energy	$E = hf = \frac{hc}{\lambda}$	E = photon energy (J); f = frequency (Hz); λ = wavelength (m); $h = 6.63 \times 10^{-34}$ J s; $c = 3.00 \times 10^8$ m s ⁻¹
Photoelectric equation	$hf = \phi + \frac{1}{2}mv_{\max}^2$; $\phi = hf_0$	f = frequency of incident radiation (Hz); ϕ = work function (J); f_0 = threshold frequency (Hz); m = electron mass (kg); $v_{\max}$ = maximum speed of photoelectrons (m s ⁻¹)
Stopping potential	$eV_s = \frac{1}{2}mv_{\max}^2 = hf - \phi$	V_s = stopping potential (V); e = electron charge (C); m = electron mass (kg); $v_{\max}$ = maximum speed of photoelectrons (m s ⁻¹); ϕ = work function (J)
Photon momentum	$p = \frac{h}{\lambda}$	p = photon momentum (kg m s ⁻¹); h = Planck constant (J s); λ = wavelength (m)
Electronvolt	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$; $E \text{ (eV)} \approx \frac{1240}{\lambda} \text{ (nm)}$	$E \text{ (eV)}$ = photon energy in electronvolts; $\lambda \text{ (nm)}$ = wavelength in nanometres

- Graph of $K_{\max}$ against f : straight line, gradient = h , intercept on the f -axis = f_0 , intercept on the energy axis = $-\phi$.

HOW TO ANSWER (MPM STYLE)

- Write $E = hf$ or $\frac{hc}{\lambda}$, convert nm to m and eV to J (or J to eV) consistently, then substitute with units.

- For photoelectric questions write $hf = \phi + K_{\max}$, then $K_{\max} = hf - \phi$; convert both energies to the same unit before subtracting.
- For explanations: state that intensity $\propto$ number of photons per second $\propto$ number of photoelectrons per second, and $K_{\max}$ depends on frequency only.
- Say 'below the threshold frequency no emission occurs, no matter how intense the light'.

SECTION A TRAPS

Trap: Increasing intensity increases the number of photoelectrons (the saturation current), not $K_{\max}$ and not the stopping potential.

Trap: Subtracting energies in different units: $hf = 4.64 \times 10^{-19} \text{ J} = 2.90 \text{ eV}$; $\phi = 2.3 \text{ eV}$ gives $K_{\max} = 0.60 \text{ eV}$. Giving 2.90 eV (forgetting ϕ) or 5.20 eV (adding ϕ) are the usual wrong options.

Trap: Threshold wavelength $\lambda_0 = \frac{hc}{\phi}$ is the LONGEST wavelength that causes emission; shorter wavelengths emit electrons. Do not reverse the inequality.

Trap: Reporting hf (in joules) when the options are in electron-volts, or stating the stopping potential in eV (it is in volts: $V_s = \frac{K_{\max}}{e}$).

TRICKS

Trick: Use $E \text{ (eV)} \approx \frac{1240}{\lambda} \text{ (nm)}$ for fast checks: 632.8 nm gives about 1.96 eV ; 450 nm gives about 2.76 eV ($4.42 \times 10^{-19} \text{ J}$).

Trick: When $K_{\max}$ is in eV, the stopping potential in volts has the same numerical value.

24.2 Wave-Particle Duality

DEFINITIONS & LAWS

- **Wave-particle duality:** radiation and matter exhibit both wave properties (diffraction, interference) and particle properties (photoelectric effect, photon momentum) depending on the experiment.
- **de Broglie hypothesis:** a moving particle of momentum p has an associated wavelength $\lambda = \frac{h}{p}$.
- Wave nature of electrons is shown by electron diffraction through a thin crystal or graphite film (Davisson-Germer experiment): a ring (diffraction) pattern is formed.
- Particle nature of light is shown by the photoelectric effect; wave nature of light by diffraction and interference.
- The wave behaviour of macroscopic objects is not observed because the de Broglie wavelength is far too small compared with any obstacle or gap.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
de Broglie wavelength	$\lambda = \frac{h}{p} = \frac{h}{mv}$	λ = de Broglie wavelength (m); h = Planck constant (J s); p = momentum (kg m s ⁻¹); m = mass (kg); v = speed (m s ⁻¹)
Wavelength from kinetic energy	$p = \sqrt{2mK}; \lambda = \frac{h}{\sqrt{2mK}}$	K = kinetic energy (J); m = mass (kg); p = momentum (kg m s ⁻¹); λ = wavelength (m)
Electron accelerated through a potential difference	$eV = \frac{1}{2}mv^2; \lambda = \frac{h}{\sqrt{2meV}}$	V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed (m s ⁻¹); λ = wavelength (m)

- Constants: $h = 6.63 \times 10^{-34}$ J s, $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg.

HOW TO ANSWER (MPM STYLE)

- Write $\lambda = \frac{h}{mv}$ (or $\frac{h}{\sqrt{2meV}}$), substitute with SI units and give the answer in metres to 3 s.f. (2 s.f. if the data are 2 s.f.).
- For evidence of both natures name the experiment AND the property shown: diffraction/interference → wave; photoelectric effect → particle.
- For an electron accelerated through V , first find the speed or momentum from $eV = \frac{1}{2}mv^2$, then $\lambda = \frac{h}{p}$.
- State: a larger momentum gives a shorter wavelength ($\lambda \propto \frac{1}{p}$).

SECTION A TRAPS

Trap: Saying that the photoelectric effect alone shows both natures: it demonstrates the particle nature only. Two different phenomena are needed.

Trap: Using the speed of light for the electron or confusing the mass of a proton (1.67×10^{-27} kg) with that of an electron. A proton at the same speed has a wavelength about 1800 times shorter.

Trap: Forgetting the square root: $\lambda = \frac{h}{\sqrt{2meV}}$, not $\frac{h}{2meV}$. For $V = 150$ V, $\lambda \approx 1.0 \times 10^{-10}$ m.

Trap: Rutherford scattering and the Millikan experiment do not show the wave nature of the electron.

TRICKS

Trick: Fast electron wavelength: λ (nm) $\approx \frac{1.23}{\sqrt{V}}$ (V in volts). For 150 V, $\lambda \approx 0.100$ nm.

Trick: Same speed, heavier particle: $\lambda \propto \frac{1}{m}$, so a proton's wavelength is $\frac{1}{1836}$ that of an electron.

24.3 Atomic Structure

DEFINITIONS & LAWS

- **Line spectrum** consists of discrete bright lines (emission) or dark lines (absorption) of definite wavelengths, characteristic of the element; it shows that electrons in atoms occupy discrete energy levels.
- **Bohr's postulates:** electrons move in certain stable orbits without radiating energy (stationary states); the angular momentum of an orbiting electron is quantised, $mvr = \frac{nh}{2\pi}$; radiation is emitted or absorbed only when an electron makes a transition between two levels, $hf = E_i - E_f$.
- **Ionisation energy** is the minimum energy needed to remove an electron from the ground state of an atom; for hydrogen it is 13.6 eV.
- **Spectral series of hydrogen:** Lyman (transitions to $n = 1$, ultraviolet), Balmer (to $n = 2$, visible), Paschen (to $n = 3$, infrared).
- **Laser action:** stimulated emission produces photons identical in phase, frequency and direction to the stimulating photon; population inversion is needed for light amplification.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Hydrogen energy levels	$E_n = -\frac{13.6}{n^2} \text{ eV}$	$n = 1, 2, 3, \dots$ ($n = 1$ is the ground state); E_n = energy of level n (eV); $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
Photon emitted or absorbed in a transition	$hf = \frac{hc}{\lambda} = E_i - E_f$	E_i = energy of the initial (higher) level (J); E_f = energy of the final level (J); f = photon frequency (Hz); λ = wavelength (m)
Rydberg formula	$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$	$R = 1.10 \times 10^7 \text{ m}^{-1}$ (Rydberg constant); λ = wavelength (m); n_i = initial level; n_f = final level
Bohr radius of orbit	$r_n = n^2 a_0$	r_n = radius of the n th orbit (m); $a_0 = 5.29 \times 10^{-11} \text{ m}$ (Bohr radius); $n = 1, 2, 3, \dots$

HOW TO ANSWER (MPM STYLE)

- Write the energies of both levels using $-\frac{13.6}{n^2}$, subtract (higher – lower) to obtain a positive photon energy, convert eV to J, then $\lambda = \frac{hc}{E}$.
- Explain line spectra in 2 points: electrons have only discrete energy levels; photons emitted in transitions have energy equal to the difference of two levels, so only specific wavelengths appear.
- Name the series (Lyman/Balmer/Paschen) and the region of the spectrum when asked.
- Give the wavelength in nm to 3 s.f. and the energy in eV.

SECTION A TRAPS

Trap: Energy levels are negative; $n = 2$ (-3.40 eV) is higher than $n = 1$ (-13.6 eV). The photon energy for $n = 2 \rightarrow 1$ is 10.2 eV, not 3.40 eV (which is the energy of the $n = 2$ level itself) or 13.6 eV (the ionisation energy).

Trap: Mistaking the Balmer wavelengths: $n = 3 \rightarrow 2$ is 658 nm (red) and $n = 5 \rightarrow 2$ is 435 nm (violet); 122 nm is the $n = 2 \rightarrow 1$ (Lyman) line, in the ultraviolet.

Trap: Writing 'electrons move in continuous orbits so any wavelength is emitted'; the quantisation of energy levels is the point.

Trap: Dividing the photon energy in eV by hc directly without converting to joules.

TRICKS

Trick: Shorter wavelength = larger energy gap. Lyman lines are shortest, then Balmer, then Paschen.

Trick: Minimum photon wavelength to ionise ground-state hydrogen: $\lambda = \frac{hc}{13.6} \text{ eV} = 91 \text{ nm}$.

24.4 X-rays

DEFINITIONS & LAWS

- X-rays** are electromagnetic waves of very short wavelength (about 10^{-11} to 10^{-9} m), produced when fast electrons are suddenly stopped by a metal target.
- Continuous spectrum** arises from the deceleration of electrons in the target (bremsstrahlung); it has a sharp cut-off at the minimum wavelength $\lambda_{\min}$, corresponding to the whole kinetic energy of an electron being converted to a single photon.
- Characteristic spectrum** consists of sharp lines produced when an incident electron ejects an inner-shell electron of the target atom and an outer electron falls to fill the vacancy; the lines are characteristic of the target material.
- Bragg's law:** constructive interference occurs for X-rays reflected from successive crystal planes when the path difference $2d \sin \theta$ is a whole number of wavelengths.
- Intensity of the X-ray beam depends on the tube current; the penetrating power (hardness) depends on the accelerating voltage.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Kinetic energy of electron at target	$eV = \frac{1}{2}mv^2$	electron reaching the target from rest; V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed at the target (m s^{-1})

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Minimum wavelength	$eV = \frac{hc}{\lambda_{\min}}; \lambda_{\min} = \frac{hc}{eV}$	$\lambda_{\min}$ = minimum wavelength (m); $h = 6.63 \times 10^{-34}$ J s; $c = 3.00 \times 10^8$ m s ⁻¹ ; $e = 1.60 \times 10^{-19}$ C; V = accelerating potential difference (V)
Bragg's law	$2d \sin \theta = n\lambda$	d = spacing of the crystal planes (m); θ = glancing angle (measured from the planes, not the normal); $n = 1, 2, 3, \dots$ (order); λ = wavelength (m)
Attenuation of X-rays	$I = I_0 e^{-\mu x}$	I_0 = incident intensity (W m ⁻²); I = transmitted intensity (W m ⁻²); μ = linear attenuation coefficient (m ⁻¹); x = thickness of the absorber (m)

HOW TO ANSWER (MPM STYLE)

- For $\lambda_{\min}$ write $\lambda_{\min} = \frac{hc}{eV}$, substitute with units; state that $\lambda_{\min}$ depends only on V and not on the target material.
- For Bragg's law, state that θ is the glancing angle between the beam and the crystal planes; write $2d \sin \theta = n\lambda$ and solve.
- For the characteristic lines, state 'transition of an outer electron to a vacancy in an inner shell'.
- When asked for the electron speed, equate eV to $\frac{1}{2}mv^2$ and convert kV to V.

SECTION A TRAPS

Trap: Using V in kV directly: $25 \text{ kV} = 2.5 \times 10^4 \text{ V}$. Forgetting this gives $\lambda_{\min}$ a factor of 1000 error (options such as 3.1×10^{-10} m for 40 kV).

Trap: Doubling V halves $\lambda_{\min}$ ($\lambda_{\min} \propto \frac{1}{V}$); options that double $\lambda_{\min}$ or leave it unchanged are wrong.

Trap: Using the angle from the normal in Bragg's law: θ is the glancing angle measured from the crystal planes.

Trap: Saying the continuous spectrum is characteristic of the target; only the sharp lines depend on the target material.

TRICKS

Trick: Quick estimate: $\lambda_{\min} \text{ (nm)} \approx \frac{1.24}{V} \text{ (kV)}$. For 25 kV, $\frac{1.24}{25} = 0.0496 \text{ nm} = 5.0 \times 10^{-11} \text{ m}$.

Trick: Bragg: $\sin \theta \leq 1$, so the wavelength must satisfy $\lambda \leq \frac{2d}{n}$ for order n .

24.5 Nanoscience

DEFINITIONS & LAWS

- **Nanoscience** is the study of materials and structures with at least one dimension of the order of 1 nm to 100 nm.
- **Quantum confinement**: when the size of a particle becomes comparable with the de Broglie wavelength of its electrons, the energy levels become discrete and their spacing increases as the size decreases.
- **Quantum dot** is a nanoscale semiconductor crystal in which electrons are confined in all three dimensions; smaller dots have a larger energy gap and emit shorter-wavelength (bluer) light.
- The surface-area-to-volume ratio of nanoparticles is very large, so a large proportion of atoms lie at the surface, which gives a high chemical reactivity and catalytic activity.
- **Scanning tunnelling microscope (STM)** uses the quantum-mechanical tunnelling current between a sharp conducting tip and the surface; the current depends very sensitively (exponentially) on the tip-surface separation, so atoms can be imaged.
- Examples: carbon nanotubes (high strength, electrical conduction), graphene, nanoparticles in sunscreens and catalysts.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Surface area to volume ratio of a sphere	$\frac{3}{r}$	r = radius of the sphere (m); ratio in m^{-1} ; the ratio increases as the size decreases
Surface area to volume ratio of a cube	$\frac{6}{L}$	L = side of the cube (m); ratio in m^{-1} ; the ratio increases as the size decreases
Particle in a one-dimensional box	$E_n = \frac{n^2 h^2}{8mL^2}$; $E \propto \frac{1}{L^2}$	E_n = energy of level n (J); $n = 1, 2, 3, \dots$; h = Planck constant (J s); m = mass of the particle (kg); L = width of the box (m)
Energy gap and wavelength	$E_{\text{gap}} = \frac{hc}{\lambda}$	E_{gap} = energy gap (J); h = Planck constant (J s); c = speed of light (m s^{-1}); λ = emitted wavelength (m); a larger gap corresponds to a shorter emitted wavelength

- $1 \text{ nm} = 10^{-9} \text{ m}$; the nanoscale is about 1–100 nm.

HOW TO ANSWER (MPM STYLE)

- For size effects state: smaller size $\rightarrow$ energy levels more widely spaced $\rightarrow$ larger bandgap $\rightarrow$ emission or absorption at a shorter wavelength.
- For reactivity state: smaller particles have a larger surface-area-to-volume ratio, so more atoms are exposed at the surface.
- For STM state: tunnelling current $\rightarrow$ very sensitive to the separation $\rightarrow$ used to map the surface atom by atom.
- Give the nanoscale range as 1 nm to 100 nm.

SECTION A TRAPS

Trap: Saying the bandgap DEcreases as the particle gets smaller; it increases, so the colour shifts towards blue (shorter wavelength).

Trap: Quoting the nanoscale as 1 μm to 100 μm or 1 pm to 100 pm; the nanoscale is 1–100 nm.

Trap: Saying that nanoparticles obey different chemistry or have fewer atoms 'so each atom is more reactive'; the reason is the large surface-area-to-volume ratio.

Trap: Attributing quantum dot colour to the number of atoms alone: bigger dots (smaller gap) emit redder light, not bluer.

TRICKS

Trick: For quantum dots: small $\rightarrow$ blue, large $\rightarrow$ red. The gap is inversely related to the wavelength emitted.

Trick: For surface-area-to-volume ratio: halving the radius doubles the ratio $\frac{3}{r}$.